

Engineering Service Learning

● Contact Point
 — Cart
 — Design (note an exact clone will also be created)

cart top
 cart base
 posts
 dashed as they are only on the outside
 lip
 wheels
 Contact Point #2
 Contact Point #1
 Original Handle
 vertical handle
 grip
 Handle Reinforcement
 Base
 Contact Point #3
 clamp
 lip

1. Assemble the apparatus shown in the figure above
 - a. Ensure the prototype is fully pushed down to maintain contact at all three points
2. Place a force plate connected to a bar against the handles so it will distribute the force equally among the two handles
3. Push against the force plate, holding the bar to the handles with friction. Stop the moment the cart begins to move.
4. Record the maximum force recorded by the force plate (F_p).

5. Move the force plate to the side of the vertical handle at the topmost position where it can still fit fully on the handle. Measure the distance (x) between the top of the original handle and the force plate to find the lever arm.
6. Push harder until the handle slips at first and record the force required to do so (F_t)
7. Remove the force plate and apparatus.
8. Gather 5 participants who are primary stakeholders in the project. Repeat the following steps for each participant until the step labeled “stop with participants.” If asked any questions related to the usage of the product, respond “I am not allowed to answer any questions, sorry.” Only do this experiment on participants who are willing to be recorded.
9. Tell the participant: “Please consider the cart on its own a 5/10 in terms of aesthetics and a 5/10 in terms of the comfort of the handle specifically. Please try it a bit so you know where this ranks for you”
 - a. Allow the participant to sample the cart
10. Say, “I will now demonstrate the assembly of this product. I will ask you to assemble it afterwards, so please pay attention.”
11. Demonstrate the assembly of the apparatus
12. Ask, “Please rank the cart in terms of its aesthetics now considering the cart without the product attached to be a 5/10.”
 - a. Record the response
13. Ask, “Please try the cart and rank the handle in terms of comfort considering the cart alone that you just tried to be a 5/10.”

- a. Allow the participant to sample the product
 - b. Record the response
14. Remove the apparatus
15. Give both handles to the person, and ask them to assemble the apparatus then disassemble it as quickly as possible on the word “start.” Ready a recording device at a flat angle so the participant’s entire body is in view.
16. Say “start” while simultaneously starting the recording.
17. Stop the recording when both handles have been removed from the cart.
18. Ask for any qualitative feedback the participant has.
 - a. Record direct quotes when possible or the general idea when not
19. Stop with participants
20. From the recording, measure the maximum angle the participant's back moves to the vertical. Record the time it took them to assemble and disassemble the apparatus (combined) from the time you said “start” to the moment the video ends.
21. Record the mass of a single handle using an electronic scale of proper size
22. Measure the smallest box the product could fit in by measuring the widest, tallest, and thickest sections of the product.
23. Add up the cost of any materials purchased

One field expert, Mr. Kiser, mentioned that a force plate would be an effective solution to objectively determine the perceived difficulty to push the cart. This logically follows because the effort required to move the card will be proportional to the force

required as a larger force requires higher effort on the person pushing the cart. We also asked for qualitative data rather than quantitative from many of the participants because we didn't want to overly restrict their ability to give feedback. Since many field experts were also the ones testing the product, we assumed their qualitative feedback on the product would turn us to better criteria to test quantitatively. For instance, Mr. Kiser noticed that the product slipped too much due to the force exerted when sharply turning the cart. Another field expert, Jared Lawson noted the same phenomenon. We considered ways to objectively measure the stability in the direction of the lip and proposed finding the maximum period of the cart before the cart began to slip to take the qualitative feedback of our field experts into consideration for future quantitative results.

We attempted to find something to quantitatively measure to objectively determine the stability of the handle. Our original idea was to spin the cart until the handle slipped, measuring the period of the cart's movement to determine how fast it moved when the cart's handle began to slip. Our field expert, Mr. Kiser, determined that this solution was quite frankly not very good because he does not intend to use the cart as a physical pendulum. He suggested instead that we measure the force required to make the handle slip at a defined lever arm. We found this to be a far better solution because it would require significantly less analysis to give us the same conclusion. As a result, we added steps 5 and 6 to objectively measure this criteria. Upon changing our methods, our field expert and primary stakeholder, Mr. Kiser signed off on the procedure.